Design note: escalation in the mean, latent dependence only where the data support it

Question. Does recorded unrest lead to later conflict (escalation), and over which time lags and distances, once episodes that drive both outcomes are accounted for?

Principles. (1) Effects of *observed* events go in the mean, as fixed effects: they are estimated directly, cheaply and with intervals. (2) The latent part only absorbs unobserved structure (confounding episodes, reporting). (3) A component enters the model only with evidence; every parameter is reported with a 95% interval. (4) Cross-dependence between the outcomes appears in the mean only; there is no coupling between latent fields.

0. Input and output

Input. Cells s (0.5°), weeks t. Outcomes $Y_{\mathrm{C}st}, Y_{\mathrm{U}st} \in \{0,1\}$: at least one recorded conflict or unrest event. Covariates $\boldsymbol{x}_{st}$, the country $c(s)$, and histories built from the outcomes (Step 2). Coarse cells $g = g(s)$ and time steps $k = k(t)$ of Δ weeks for the space-time fields.
Output. The escalation kernels $\kappa_{\mathrm{U}\to\mathrm{C}}(b,r)$ and $\kappa_{\mathrm{C}\to\mathrm{U}}(b,r)$ with intervals; the loading λ with its interval; maps of the latent fields; predictive probabilities for held-out data.

1. Data level

Given the linear predictors, $Y_{\mathrm{C}st}$ and $Y_{\mathrm{U}st}$ are independent Bernoulli variables, $\operatorname{logit} P(Y_{jst} = 1) = \eta_{jst}$ for $j \in \{\mathrm{C},\mathrm{U}\}$, independent over cells and weeks. All dependence between the outcomes comes from the histories (Step 2) and from the loading λ (Step 3).

2. Linear predictors

$$\eta_{\mathrm{C}st} = \boldsymbol{x}_{st}^{\top}\beta_{\mathrm{C}} + \delta_{\mathrm{C}}\, h_{\mathrm{C}st} + \delta_{\mathrm{C}}^{nb}\, \bar{h}_{\mathrm{C}st} + \textstyle\sum_{b,r} \kappa_{\mathrm{U}\to\mathrm{C}}(b,r)\, H_{\mathrm{U}st}(b,r) + u_{\mathrm{C},c(s)} + e_{\mathrm{C}s} + w_{\mathrm{C}s} + v_{\mathrm{C}gk},$$
$$\eta_{\mathrm{U}st} = \boldsymbol{x}_{st}^{\top}\beta_{\mathrm{U}} + \delta_{\mathrm{U}}\, h_{\mathrm{U}st} + \delta_{\mathrm{U}}^{nb}\, \bar{h}_{\mathrm{U}st} + \textstyle\sum_{b,r} \kappa_{\mathrm{C}\to\mathrm{U}}(b,r)\, H_{\mathrm{C}st}(b,r) + u_{\mathrm{U},c(s)} + e_{\mathrm{U}s} + w_{\mathrm{U}s} + \lambda\, v_{\mathrm{C}gk} + v_{\mathrm{U}gk}.$$

Own histories (as now). $h_{jst} = \log(1 + \text{events of type } j \text{ in cell } s \text{ in weeks } t-4,\ldots,t-1)$, and $\bar{h}_{jst}$ is its mean over the rook neighbours of s.

Escalation kernel (new). The history of the other outcome is split by lag band b and distance ring r. For unrest,

$$H_{\mathrm{U}st}(b,r) = \text{mean over cells } s' \in R_r(s) \text{ of } \log\bigl(1 + \text{unrest events in } s' \text{ in the weeks } t - B_b\bigr),$$

with lag bands $B_1 = \{1,\ldots,4\}$, $B_2 = \{5,\ldots,13\}$ and $B_3 = \{14,\ldots,26\}$ weeks before t, and rings $R_0(s) = \{s\}$, $R_1(s)$ = rook neighbours and $R_2(s)$ = cells 50–300 km away. $H_{\mathrm{C}st}(b,r)$ is the same for conflict. All are standardised on the training rows. This gives 9 coefficients per direction. The current model is the special case $b = 1$, $r \le 1$: $\kappa_{\mathrm{U}\to\mathrm{C}} = \kappa_{\mathrm{U}\to\mathrm{C}}(1,0)$ and $\kappa_{\mathrm{U}\to\mathrm{C}}^{nb} = \kappa_{\mathrm{U}\to\mathrm{C}}(1,1)$. All fixed effects have $\mathcal{N}(0,10^4)$ priors.

3. Latent level

Static random effects, one independent set per outcome j: $u_{j,c} \sim \mathcal{N}(0,\sigma_{u,j}^2)$ (country), $e_{js} \sim \mathcal{N}(0,\sigma_{e,j}^2)$ (cell, no spatial correlation) and w_j, a Matérn field with sea barriers (regional, constant in time).

Space-time fields. Two fields, independent of each other: v_{C} (the conflict field) and v_{U} (unrest only). Each is an AR(1) in time with Matérn innovations with sea barriers. For coarse cell g and step k,

$$v_{jgk} = \varphi_j\, v_{jg,k-1} + \sqrt{1-\varphi_j^2}\;\varepsilon_{jgk}, \qquad (\varepsilon_{j1k},\ldots,\varepsilon_{jGk}) \sim \text{Matérn}(\rho_j,\sigma_j),$$

independent over k. The only latent link between the outcomes is the loading λ: the conflict field also enters unrest. With $\lambda = 0$ the model is two separate univariate models. This is the current extended model, renamed: v_{C} is its "shared" field.

4. Parameters

Fixed effects: β_j, δ_j, δ_j^{nb}, and 2×9 kernel coefficients. Hyperparameters (15, as now): per outcome $\sigma_{u,j}, \sigma_{e,j}, \rho_{w,j}, \sigma_{w,j}$; per field $\rho_j, \sigma_j, \varphi_j$; and $\lambda \sim \mathcal{N}(0, 2^2)$. PC priors as in the paper, and $\operatorname{atanh} \varphi_j \sim \mathcal{N}(0, 1.5^2)$. Inference: Laplace empirical Bayes as now, with intervals for every hyperparameter from the Hessian.

5. Evidence so far (Africa and the Middle East)

component	evidence	status
space-time field, conflict	log-ML +2,421 over static terms only	supported
space-time field, unrest	log-ML +1,592 over static terms only	supported
latent link λ	$\hat{\lambda} = 0.52$; log-ML +129 over two independent fields	interval pending
	held-out joint log score 0.11387, against 0.11422 (independent fields) and 0.11325 (static only)	mixed
one shared field only	log-ML −575 against two independent fields	rejected
$\kappa_{\mathrm{U}\to\mathrm{C}}$ (4 weeks, same cell)	0.094 [0.065, 0.123]	supported
$\kappa_{\mathrm{C}\to\mathrm{U}}$ (4 weeks, same cell)	0.069 [0.042, 0.096]	supported

The evidence for latent dependence between the outcomes rests on one parameter, λ. *Decision rule:* if the interval for λ excludes zero, keep it; if not, drop it and fit the two outcomes separately, with escalation in the mean only.

6. What is not in the model, and what would bring it in

No conflict-only third field and no latent coupling between fields. They enter only if the fitted model leaves lagged cross-dependence unexplained. *Check* (cheap, no refit): lagged cross-correlations between conflict and unrest Pearson residuals, aggregated to coarse cells and steps, at lags of $0, 1, 2, \ldots$ steps. If they lie within the band expected under the fitted model (from posterior predictive draws), the simple model is enough.

7. Identification and checks

The lag bands B_2 and B_3 overlap the time scale of the fields ($\Delta = 13$ or 26 weeks), so long-lag escalation competes with the persistence φ_{C}. Simulation (the S2 design, 20 replicates each): (i) shared episodes only, where all long-lag coefficients should be near zero; (ii) escalation at band B_2 and ring R_2 only, which should be recovered with correct coverage. Cost: kernel features and simulation, about 2 days; refits as now.